

A close-up photograph of a hand holding a blue ink stamp. The stamp is positioned over a document, and the hand is in the process of pressing it down. The background is dark and out of focus.

LOAN APPROVAL PREDICTION

**USING MACHINE LEARNING
TECHNIQUES**

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ABSTRACT

Loan approval prediction is an important process in the banking and financial sector because banks and financial institutions must carefully evaluate applicants before granting loans. Traditional loan approval methods can be time-consuming and may sometimes produce inconsistent decisions. Machine learning techniques help improve this process by providing faster, automated and data-driven loan approval predictions.

This study worked with a publicly available dataset from Kaggle which represents loan application records from Dream Housing Finance, that contained information on 614 applicants with 13 variables such as gender, marital status, education, income, loan amount, credit history, property area and loan approval status.

Several preprocessing techniques were applied before model development, including handling missing values, encoding categorical variables, feature scaling and exploratory data analysis (EDA). Multiple supervised machine learning models were then developed and evaluated, including Logistic Regression, Decision Tree, Random Forest, Gradient Boosting and Support Vector Machine (SVM).

Model performance was evaluated using Accuracy, F1-Score, ROC-AUC score, confusion matrix analysis, classification reports and ROC curve analysis.

Among all evaluated models, Logistic Regression demonstrated the best overall performance with an accuracy of 86.18%, F1-Score of 0.908 and ROC-AUC score of 0.848. Although Decision Tree achieved a similar accuracy value, Logistic Regression showed better classification balance, stronger discrimination capability and higher interpretability.

Feature importance analysis revealed that Credit_History was the most influential variable affecting loan approval decisions, followed by variables such as property area, marital status, education and income-related features.

The findings obtained from exploratory data analysis were strongly supported by the advanced machine learning analysis results, increasing the reliability and validity of the study. Compared to several previous studies that mainly focused on single models or relied heavily on resampling techniques such as SMOTE, this study provides a broader comparative analysis of multiple machine learning algorithms while demonstrating that strong predictive performance can still be achieved without applying complex balancing techniques when class imbalance is moderate.

Overall, this study demonstrates the practical usefulness of machine learning techniques for automated loan approval prediction. The proposed approach can help financial institutions improve decision-making consistency, reduce manual effort, support financial risk management and provide faster and more reliable loan approval processes.

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1. INTRODUCTION

Loan approval is an important process in the banking and financial sector because financial institutions must carefully evaluate applicants before granting loans. The decision to approve or reject a loan is usually based on several demographic, financial and social factors such as income level, education, credit history, employment status and property area. Accurate loan approval decisions are essential for reducing financial risk and ensuring that loans are provided to eligible applicants.

The dataset used in this study was obtained from Kaggle and it represent loan application records from Dream Housing Finance, a housing finance company that provides home loans to customers across urban, semi-urban and rural areas. Customers first apply for a home loan, after which the company evaluates their eligibility before making approval decisions. Traditional loan approval processes are often manual, time-consuming and may sometimes lead to inconsistent or biased decisions. As the number of loan applications increases, there is a growing need for automated and data-driven decision-making systems.

Machine learning techniques provide an effective solution for improving loan approval prediction. By analyzing historical applicant data, machine learning models can identify important patterns and relationships that influence loan approval decisions. These models can support financial institutions by improving prediction accuracy, reducing manual effort, increasing consistency and supporting faster decision-making.

This study focuses on developing a machine learning-based system to predict loan approval status using applicant information. Several supervised machine learning models were developed and compared to identify the most suitable model for loan approval prediction. The study also aims to identify the most important variables influencing loan eligibility through feature importance analysis.

Objectives of the study:

The main objective of this study is to develop a machine learning model capable of accurately predicting loan approval status.

The specific objectives include identifying the most important variables influencing loan eligibility, evaluating model performance using accuracy and other classification metrics, as well as improving the efficiency and consistency of the loan approval process.

This study is significant because it demonstrates the practical application of machine learning techniques in banking and financial systems. And also it can help to automate the loan approval process, reduce manual workload, support better financial risk management and assist institutions in making more consistent and reliable lending decisions.

2. LITERATURE REVIEW

Predicting loan approval using machine learning has gained significant attention in the banking and financial sector, as it helps minimize financial risk and supports more efficient decision-making. Various studies have explored the performance and applicability of different machine learning and statistical techniques for this purpose.

The study by Sinap, V. (2024). A comparative study of loan approval prediction using machine learning methods. *Gazi Üniversitesi Fen Bilimleri Dergisi Part C: Tasarım ve Teknoloji*, 12(2), 644–663. compared several machine learning algorithms for loan approval prediction and the study reported that Random Forest achieved the highest prediction accuracy (97.71%), while Logistic Regression remained highly useful because of its interpretability and transparency in financial applications. The findings also showed that variables such as marital status, income level, education, and gender significantly influenced loan approval decisions. However, the study used limited model evaluation techniques like K-fold cross validation and focused mainly on prediction accuracy.

A recent study by Velvigneswar, J., & Kumari, P. S. (2025). Loan approval prediction using machine learning. *International Journal of Advanced Research in Computer and Communication Engineering*, 14(7). developed a machine learning-based loan approval prediction system using algorithms such as Logistic Regression, Random Forest, Decision Tree and SVM. The study applied preprocessing techniques including missing value imputation, encoding, feature scaling and SMOTE for class balancing. The results showed that SVM achieved the best overall performance among the evaluated models with an accuracy of 79.67%, while others achieved comparatively lower performance levels. However, despite applying SMOTE for balancing the dataset, the achieved accuracy values remained lower compared to several recent studies. The research mainly focused on prediction performance and provided limited discussion regarding detailed model adequacy testing and interpretability analysis.

Another related study by Jadhav, A., et al. (2023). Loan approval prediction using machine learning. *International Research Journal of Modernization in Engineering Technology and Science*, 5(5). investigated loan approval prediction using several machine learning algorithms, including Logistic Regression, Decision Tree, Random Forest and Gradient Boosting. The study used applicant demographic and financial information to predict loan approval status. Their results showed that Logistic Regression achieved the highest prediction accuracy of 80%, outperforming the other evaluated models. However, the authors also noted that prediction performance heavily depends on the quality and quantity of available data. In addition, the study mainly focused on accuracy-based evaluation and provided limited discussion regarding feature importance analysis and detailed model validation techniques.

Overall, previous studies show that machine learning is effective for loan approval prediction. However, many of them used only a few models, focused mainly on accuracy or had limited interpretability analysis. Some studies also used balancing techniques such as SMOTE or undersampling without fully examining whether class imbalance significantly affected model performance. Therefore, this study compares multiple machine learning models using evaluation metrics such as Accuracy, F1-Score and ROC-AUC and also includes feature importance analysis to identify key factors influencing loan approval decisions

3. THEORY AND METHODOLOGY

This section outlines the methodology used to develop the predictive model for loan approval prediction. The study adopted a data-driven approach using supervised machine learning techniques to analyze demographic, financial and social characteristics of loan applicants. The dataset used in this research was obtained from Kaggle website and contains information related to loan applicants along with the loan approval status.

The analysis, preprocessing, visualization and model development processes were performed using Python and libraries such as pandas, NumPy, scikit-learn, matplotlib and seaborn.

3.1 DATA PREPROCESSING

Several preprocessing techniques were applied before model development.

1. Handling Missing Values

The dataset was checked for missing values and duplicated records. Imputation techniques were used instead of removing records in order to preserve the dataset size and avoid unnecessary loss of information. Missing values in categorical variables were handled using mode imputation and missing values in numerical variables were handled using median imputation because the variables showed skewed distributions. It was selected because it is less sensitive to extreme values and outliers compared to mean imputation.

2. Outlier Detection

Outliers were identified in variables such as ApplicantIncome, CoapplicantIncome and LoanAmount using the Interquartile Range (IQR) method.

$$\text{IQR} = Q3 - Q1$$

$$\text{Lower bound} = Q1 - 1.5 * \text{IQR}$$

$$\text{Upper bound} = Q3 + 1.5 * \text{IQR}$$

Data points beyond the lower bound and upper bound are considered as outliers.

However, outliers were not removed, as they may represent genuine financial differences among applicants and could contain important predictive information for loan approval.

3. Encoding Categorical Variables

Machine learning models require numerical input variables.

- Label encoding

Categorical variables such as binary variables were converted into numerical form and Dependents variable encoded after converting categories such as 3+ into numerical form to maintain consistency within the dataset.

- One-Hot Encoding

Property_Area variable was transformed using One-Hot Encoding to avoid introducing ordinal relationships among categories.

4. Feature Scaling

Feature scaling was performed using StandardScaler to standardize numerical variables before training models such as Logistic Regression and SVM.

Standardization transforms variables to have zero mean and unit variance. This ensures all the variables are in same scale.

The standardization formula is:

$$Z = \frac{X - \mu}{\sigma}$$

Where:

- X = original value
- μ = mean of the variable
- σ = standard deviation

3.2 EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) was conducted to understand the structure, distribution and relationships within the dataset before model development.

Various visualizations such as histograms, boxplots, countplots, scatterplots, pie charts and heatmaps were used to identify variable distributions, outliers, missing value patterns, class imbalance and relationships between variables and loan approval status.

3.3 TRAIN-TEST SPLIT

The dataset was divided into training and testing sets using an 80:20 ratio through the `train_test_split()` function.

The training dataset was used for model learning, while the testing dataset was used to evaluate predictive performance on unseen data.

3.4 MODEL DEVELOPMENT

Five supervised machine learning models were developed and evaluated:

1. Logistic Regression

Logistic Regression is a supervised machine learning algorithm commonly used for binary classification problems. It predicts the probability of an observation belonging to a particular class. In this study, Logistic Regression was used to predict whether a loan application would be approved (Y) or rejected (N).

The model uses the logistic (sigmoid) function to generate probability values between 0 and 1.

$$P(Y = 1) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n)}}$$

Where:

- $P(Y = 1)$ = probability of loan approval
- β_0 = intercept
- $\beta_1, \beta_2, \dots, \beta_n$ = regression coefficients
- $X_1, X_2, \dots, X_n$ = predictor variables

2. Decision Tree (DT)

Decision Tree is a supervised learning algorithm that splits the dataset into smaller groups based on feature values. The model forms a tree-like structure containing decision nodes and leaf nodes representing prediction outcomes.

The Gini Index used for splitting is:

$$Gini = 1 - \sum p_i^2$$

Where:

- p_i = probability of a class within a node

Decision Trees can model nonlinear relationships and are easy to interpret. However, they may become prone to overfitting if the tree grows too deep.

3. Random Forest (RF)

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting.

The Random Forest prediction is obtained using majority voting:

$$\hat{y} = \text{mode}(T_1(x), T_2(x), \dots, T_n(x))$$

Where:

- $T_1, T_2, \dots, T_n$ represent individual decision trees
- $\hat{y}$ = final predicted class

4. Gradient Boosting

Gradient Boosting is an ensemble technique that builds models sequentially, where each new model attempts to correct the errors of previous models. This mode is effective for capturing complex relationships within structured datasets.

The general boosting equation is:

$$F_m(x) = F_{m-1}(x) + h_m(x)$$

Where:

- $F_m(x)$ = updated prediction model
- $h_m(x)$ = weak learner added at iteration m

5. Support Vector Machine (SVM)

Support Vector Machine (SVM) is a supervised classification algorithm that identifies the optimal hyperplane separating different classes.

The linear SVM equation is $w \cdot x + b = 0$

Where:

- w = weight vector
- x = feature vector
- b = bias term

4. DATA DESCRIPTION

The dataset used in this study is the **Loan Prediction Dataset** publicly available on Kaggle. It contains information on 614 loan applicants, described by 13 demographic, financial and social variables related to the loan application process.

The dataset includes both numerical and categorical features such as applicant income, loan amount, education level, credit history and property area.

In addition, it contains a binary target variable indicating whether the loan application was approved (Y) or not approved (N).

4.1 VARIABLES AND DESCRIPTIONS

Table 4.1

Description of Variables

Variable	Description	Type
1.Loan_ID	Unique Loan ID	Categorical
2.Gender	Gender of the applicant (Male/ Female)	Categorical
3.Married	Marital status of the applicant (Y= Yes/ N= No)	Categorical
4.Dependents	Number of dependents that applicant have	Categorical
5.Education	Educational qualification of the applicant (Graduate/ Under Graduate)	Categorical
6.Self_Employed	Whether the applicant is self-employed (Y= Yes/ N= No)	Categorical
7.ApplicantIncome	Monthly income of the applicant	Numerical
8.CoapplicantIncome	Monthly income of the co-applicant (if any)	Numerical
9.LoanAmount	Loan amount requested (in thousands)	Numerical
10.Loan_Amount_Term	Term of the loan (in months)	Numerical
11.Credit_History	Credit history meets guidelines (1= Yes/ 0= No)	Binary
12.Property_Area	Type of area where the applicant lives (Urban/ Semi-Urban/ Rural)	Categorical
13.Loan_Status	(Target) Loan approved or not (Y= Yes/ N= No)	Categorical

- A snapshot of the dataset is shown below to provide an overview of the raw data structure. It includes key variables used in the analysis before preprocessing and modeling.

Figure 4.1

Structure of Dataset

	A	B	C	D	E	F	G	H	I	J	K	L	M
	Loan_ID	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	Credit_History	Property_Area	Loan_Status
1	LP001002	Male	No	0	Graduate	No	5849	0		360		1 Urban	Y
2	LP001003	Male	Yes	1	Graduate	No	4583	1508	128	360		1 Rural	N
3	LP001005	Male	Yes	0	Graduate	Yes	3000	0	66	360		1 Urban	Y
4	LP001006	Male	Yes	0	Not Graduate	No	2583	2358	120	360		1 Urban	Y
5	LP001008	Male	No	0	Graduate	No	6000	0	141	360		1 Urban	Y
6	LP001011	Male	Yes	2	Graduate	Yes	5417	4196	267	360		1 Urban	Y
7	LP001013	Male	Yes	0	Not Graduate	No	2333	1516	95	360		1 Urban	Y
8	LP001014	Male	Yes	3+	Graduate	No	3036	2504	158	360		0 Semiurban	N
9	LP001018	Male	Yes	2	Graduate	No	4006	1526	168	360		1 Urban	Y
10	LP001020	Male	Yes	1	Graduate	No	12841	10968	349	360		1 Semiurban	N
11	LP001024	Male	Yes	2	Graduate	No	3200	700	70	360		1 Urban	Y
12	LP001027	Male	Yes	2	Graduate	No	2500	1840	109	360		1 Urban	Y
13	LP001028	Male	Yes	2	Graduate	No	3073	8106	200	360		1 Urban	Y
14	LP001029	Male	No	0	Graduate	No	1853	2840	114	360		1 Rural	N

4.2 DATA PREPROCESSING

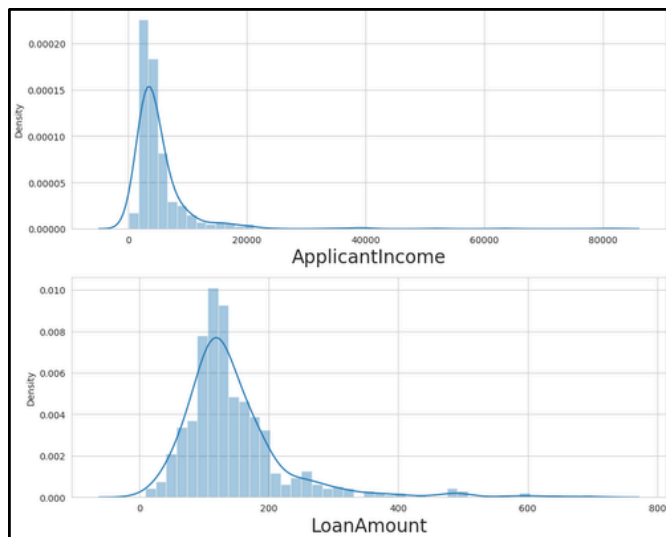
Data preprocessing was carried out to improve the quality of the dataset and prepare it for machine learning analysis. Several preprocessing techniques were applied before model development.

- Handling Missing Values:

Missing values were identified in variables such as Gender, Married, Dependents, Self_Employed, LoanAmount, Loan_Amount_Term and Credit_History. Before imputing numerical variables, their distributions were examined using histograms to identify skewness and understand the data pattern.

Figure 4.2

Distribution of Numerical Variables



As shown in Figure 4.2, since variables such as ApplicantIncome and LoanAmount showed skewed distributions, median imputation was used for numerical variables to reduce the effect of extreme values and outliers.

Mode imputation was applied for categorical variables.

- Variable Transformation:

The Dependents variable contained a category labeled 3+, which was transformed into a numerical value for analysis. Binary categorical variables such as Gender, Married, Education, Self_Employed and Loan_Status were converted into numerical form using Label Encoding.

- Handling Categorical Variables:

The Property_Area variable contained multiple unordered categories (Urban, Semiurban and Rural). Therefore, One-Hot Encoding was applied to transform the variable into separate binary columns suitable for machine learning models.

- Outlier Detection:

Outliers were identified in variables such as ApplicantIncome, CoapplicantIncome and LoanAmount using boxplots and the Interquartile Range (IQR) method. However, outliers were not removed, as they may represent genuine financial differences among applicants and could contain important predictive information for loan approval.

- Feature Engineering:

A new variable named TotalIncome was created by combining ApplicantIncome and CoapplicantIncome to better represent the overall income level of loan applicants.

- Feature Scaling:

StandardScaler was applied to numerical variables to standardize the data before model training. This step was particularly important for algorithms such as Logistic Regression and Support Vector Machine (SVM), which are sensitive to feature scale.

- Train-Test Split:

The dataset was divided into training (80%) and testing (20%) sets using train_test_split to evaluate model performance on unseen data and ensure a fair assessment of the models.

5. EXPLORATORY DATA ANALYSIS

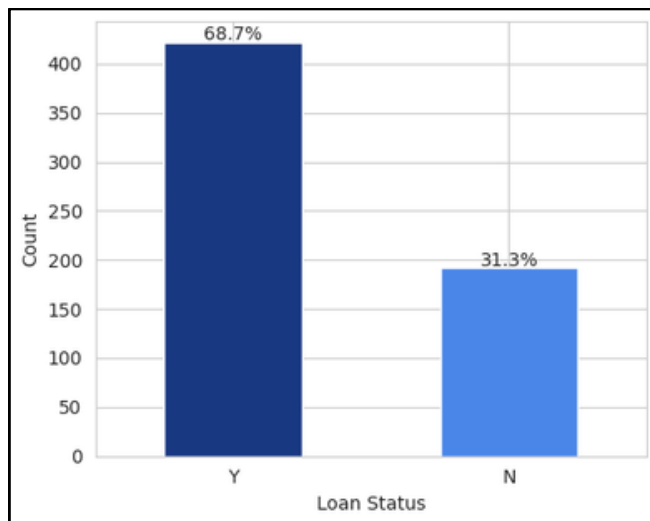
Exploratory Data Analysis (EDA) was conducted to understand the structure and characteristics of the dataset before model development. The analysis included identifying outliers, examining variable distributions and exploring relationships between variables and loan approval status.

Here various visualizations such as bar charts, histograms, boxplots, scatterplots and correlation heatmaps were used to identify important patterns and trends within the data

5.1 UNIVARIATE ANALYSIS

Figure 5.1

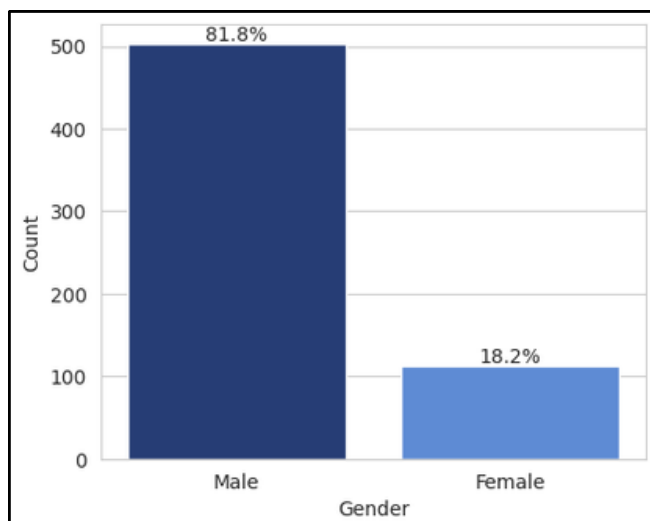
Distribution of Loan Status



The Figure 5.1 illustrates the distribution of loan approval status among applicants. A larger proportion of applicants were approved for loans compared to rejected applicants. Although the dataset shows a slight class imbalance, the difference is not severe and therefore no resampling techniques were applied during model development.

Figure 5.2

Distribution of Gender



In Figure 5.2 ,the gender distribution shows that the majority of loan applicants were male, while a smaller proportion were female applicants. This indicates that male applicants are more represented in the dataset, which may reflect the demographic characteristics of loan applicants in the selected financial context.

Figure 5.3

Distribution of Dependents

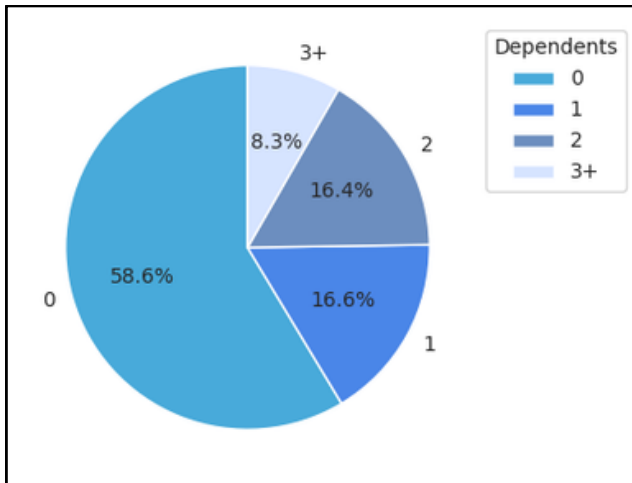
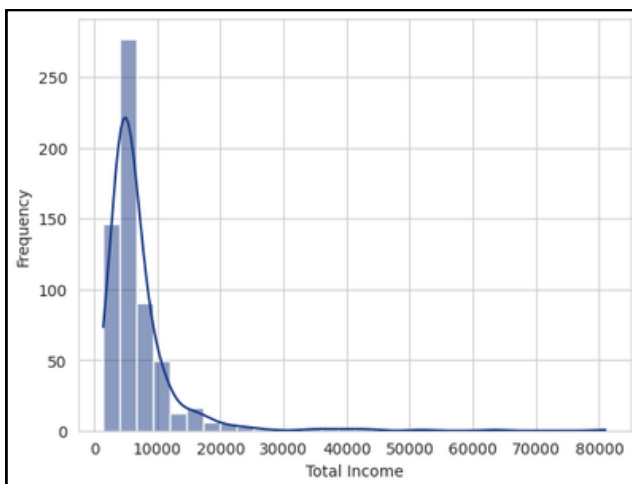


Figure 5.3 illustrates the distribution of dependents and reveals that most applicants had either zero or one dependent, while fewer applicants had larger family responsibilities. This pattern suggests that smaller family sizes are more common among loan applicants within the dataset.

Figure 5.4

Distribution of Applicant Total Income



The Figure 5.4, histogram of total applicant income demonstrates a positively skewed distribution, where the majority of applicants fall within lower to moderate income ranges, while a small number of applicants have significantly higher incomes.

Figure 5.5

Distribution of Loan Amount Using Boxplot

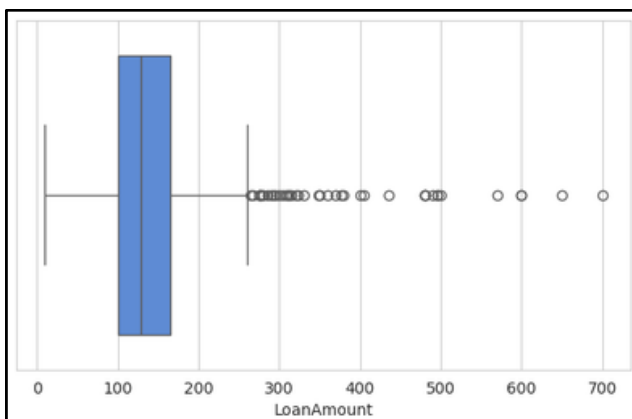
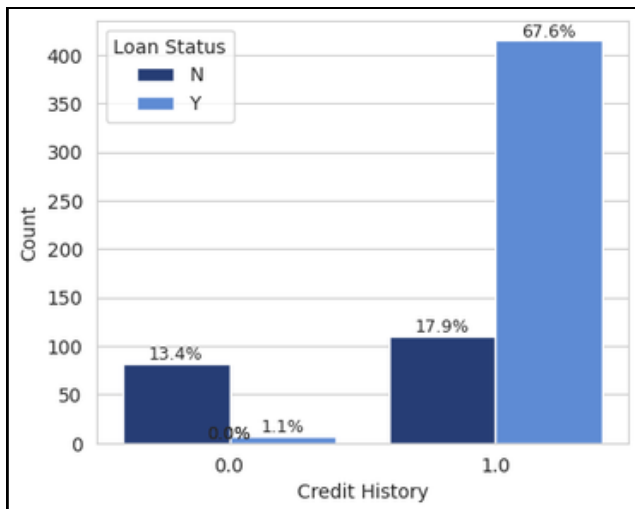


Figure 5.5 reveals the presence of several outliers in the loan amount variable. These outliers indicate that a small number of applicants requested substantially larger loan amounts compared to the majority of applicants. However, these values were retained as they likely reflect genuine financial variation important for loan prediction.

5.2 BIVARIATE ANALYSIS

Figure 5.6

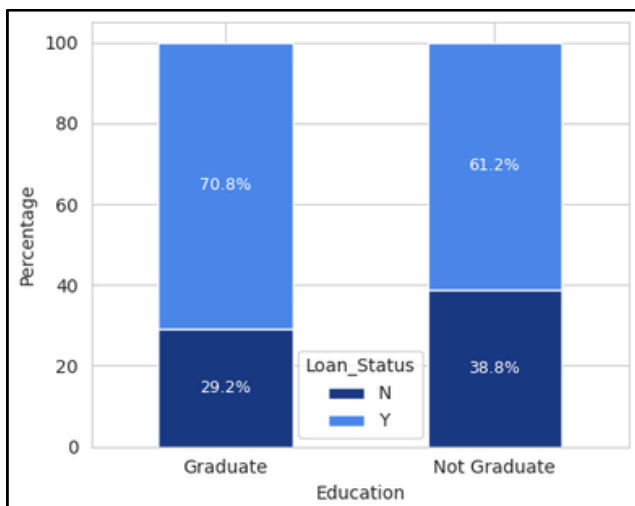
Distribution of Loan Status Vs Credit History



The relationship between credit history and loan approval status indicates a strong association between the two variables. In Figure 5.6 it shows applicants with a valid credit history were considerably more likely to receive loan approval compared to those without a valid credit history. This suggests that credit history is one of the most influential factors in the loan approval process.

Figure 5.7

Distribution of Education Vs Loan Status



The analysis of education level and loan approval status (Figure 5.7) shows that graduate applicants had a slightly higher proportion of approved loans (70.8%) compared to non-graduate applicants (61.2%). This may indicate that education level contributes to the financial credibility of applicants during loan evaluation.

Figure 5.8

Distribution of Marital Status Vs Loan Status

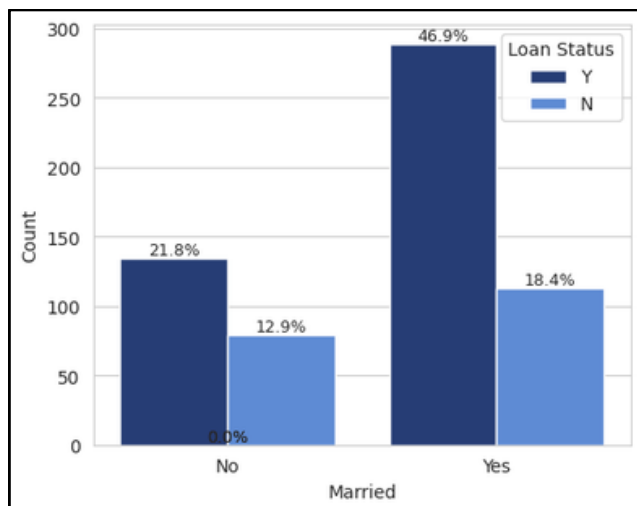


Figure 5.8 shows the relationship between marital status and loan approval status. Married applicants showed a noticeably higher loan approval percentage (46.9%) compared to unmarried applicants. In contrast, unmarried applicants had relatively lower approval rates and slightly higher rejection proportions. This indicates that marital status may have a meaningful relationship with loan approval decisions.

Figure 5.9

Distribution of Property Area Vs Loan Status

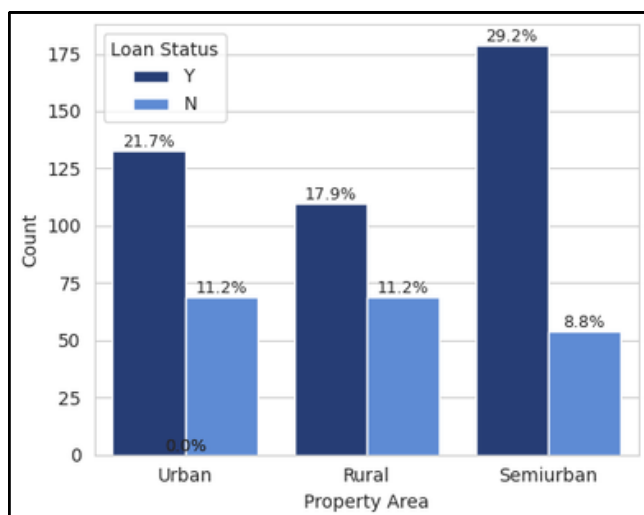


Figure 5.9 illustrate the relationship between property area and loan approval status. Applicants from Semiurban areas recorded the highest loan approval percentage (29.2%) compared to Urban and Rural areas. Rural applicants showed comparatively lower approval rates, while rejection rates were relatively higher among Urban and Rural applicants. These findings suggest that property area may influence loan approval status.

Figure 5.10

Distribution of Applicant Income Vs Loan Amount

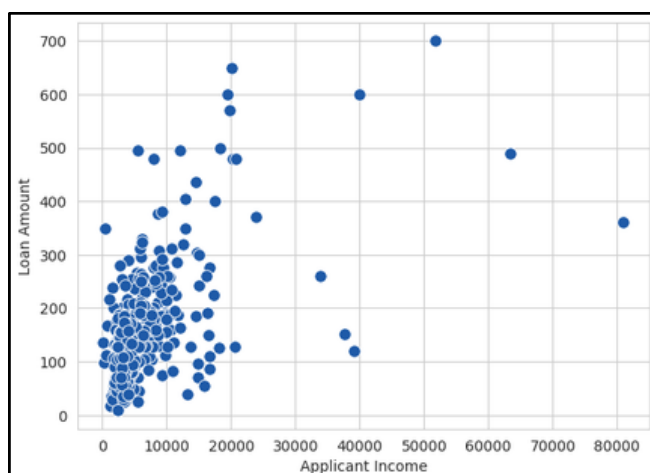


Figure 5.10 shows a moderate positive relationship, where applicants with higher incomes generally tend to apply for larger loan amounts. Most observations are concentrated within the lower income and lower loan amount region, while a few extreme values appear at higher income levels. The increasing spread of loan amounts with higher incomes suggests slight heteroscedasticity. This indicates that additional factors may also influence loan approval decisions.

5.3 MULTIVARIATE ANALYSIS

Figure 5.11

Correlation Heatmap of Loan Dataset Variables

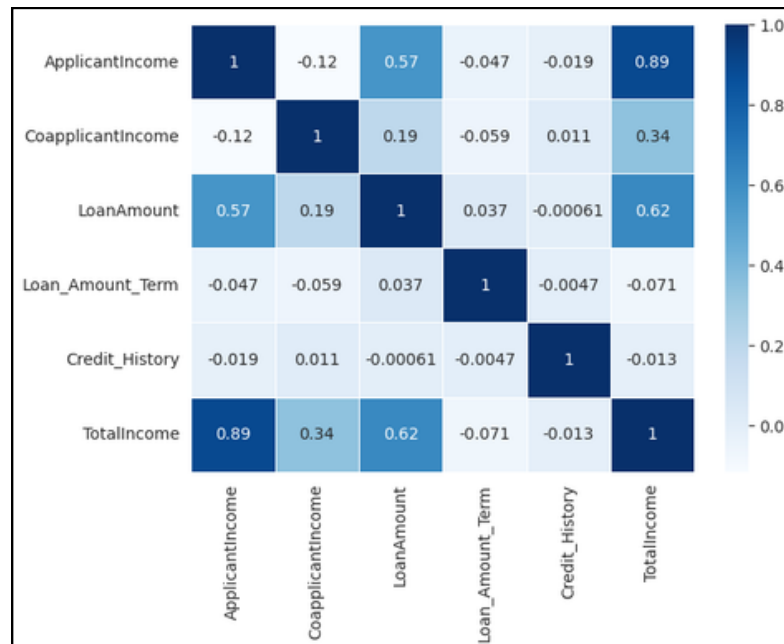


Figure 5.11 shows the correlation heatmap of numerical variables in the dataset. A stronger positive relationship is observed between applicant income, coapplicant income and total income. Most other variables show weak to moderate correlations, indicating low multicollinearity in the dataset.

- Overall, the exploratory data analysis provided useful insights into the dataset. It identified skewed data, outliers and relationships between variables and loan approval status. Credit history was found to be the most important factor, while income, education and property area also showed patterns.
- The EDA results helped guide data preprocessing steps such as handling missing values, encoding, feature engineering and scaling, which supported building effective machine learning models for loan prediction.

6. ADVANCED ANALYSIS

Multiple machine learning models were evaluated to identify the most suitable algorithm for loan approval prediction. Five supervised machine learning models were developed and evaluated to predict loan approval status using applicant demographic, financial and social information. The models used in this study were Logistic Regression, Decision Tree, Random Forest, Gradient Boosting and Support Vector Machine (SVM). Hyperparameter tuning was performed using GridSearchCV to identify the optimal parameter combinations for each model.

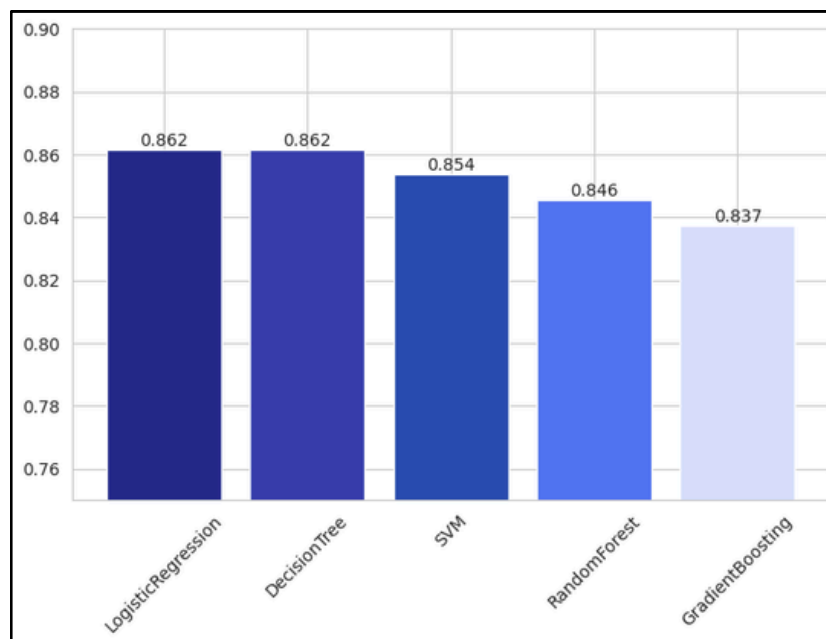
The primary objective of these models was to classify whether a loan application would be approved (Loan_Status = Y) or rejected (Loan_Status = N). Model performance was evaluated using Accuracy, F1-Score, ROC-AUC score, Confusion Matrix, Classification Report and Feature Importance Analysis. These evaluation metrics provided a comprehensive understanding of model performance and predictive capability.

6.1 MODEL PERFORMANCE

To evaluate the effectiveness of different machine learning models for loan approval prediction, several evaluation metrics were considered, including Accuracy, F1-Score and ROC-AUC score. These metrics were selected because the dataset showed a slight class imbalance, making it important to assess not only accuracy but also the balance between precision and recall.

Figure 6.1

Accuracy Comparison of Machine Learning Models



- The model comparison results indicate that Logistic Regression and Decision Tree achieved the highest classification accuracy of 86.18%. Among all evaluated models, Logistic Regression demonstrated the best overall performance with an F1-Score of 0.908 and an ROC-AUC score of 0.848. These findings suggest that Logistic Regression was highly effective in distinguishing between approved and rejected loan applications.
- Support Vector Machine (SVM) also showed strong predictive performance, achieving an accuracy of 85.37% and an F1-Score of 0.903. However, its ROC-AUC score was comparatively lower (0.714), indicating weaker class separation ability compared to Logistic Regression.
- Tree-based ensemble methods such as Random Forest and Gradient Boosting achieved slightly lower accuracy scores of 84.55% and 83.74% respectively. Although these models performed reasonably well, they did not outperform Logistic Regression in terms of overall classification performance.

Overall, the results suggest that Logistic Regression provided the best balance between predictive accuracy, classification stability, interpretability and discrimination ability for this dataset.

6.2 MODEL COMPARISON

Table 6.1

Performance Comparison of Machine Learning Models

Model	Accuracy	F1-score	AUC
Logistic Regression	0.861789	0.908108	0.847678
Decision Tree	0.861789	0.907104	0.783746
SVM	0.853659	0.903226	0.713932
Random Forest	0.845528	0.897297	0.838390
Gradient Boosting	0.837398	0.891304	0.816718

- After comparing all evaluated models, Logistic Regression was selected as the final model because it achieved the best overall balance between Accuracy, F1-Score, ROC-AUC performance and interpretability.
- Although Decision Tree achieved similar accuracy, Logistic Regression demonstrated slightly better classification stability and discrimination capability, making it more suitable for loan approval prediction.

6.3 MODEL EVALUATION AND VALIDATION

Several evaluation techniques were used to assess the performance and reliability of the selected Logistic Regression model, including the classification report, confusion matrix, ROC curve and feature importance analysis.

6.3.1 Classification report

Figure 6.2
Classification report of Logistic Regression Model

Classification Report:				
	precision	recall	f1-score	support
0	0.96	0.58	0.72	38
1	0.84	0.99	0.91	85
accuracy			0.86	123
macro avg	0.90	0.78	0.81	123
weighted avg	0.88	0.86	0.85	123

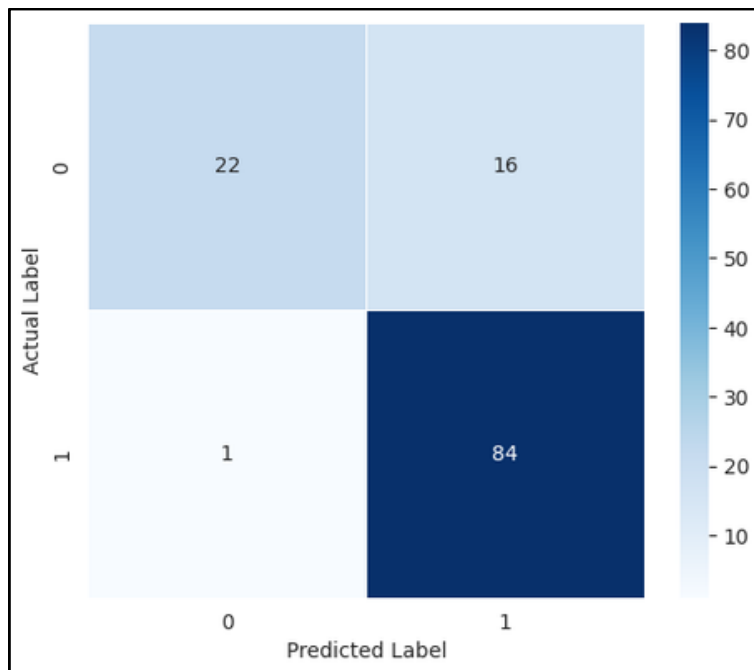
- The classification report showed that the Logistic Regression model performed well in predicting loan approval status. For approved loan applications (Class 1), the model achieved a precision of 0.84, recall of 0.99 and an F1-Score of 0.91, indicating strong ability in correctly identifying approved applicants.
- For rejected loan applications (Class 0), the model achieved a precision of 0.96 and a recall of 0.58. This suggests that although some rejected applications were incorrectly predicted as approved, the model was still effective in identifying rejected applicants with high precision.

Overall, the model achieved an accuracy of 86% and a weighted average F1-score of 0.85, demonstrating strong predictive performance and classification reliability.

6.3.2 Confusion Matrix

Figure 6.3

Confusion Matrix of Logistic Regression Model(Best Model)

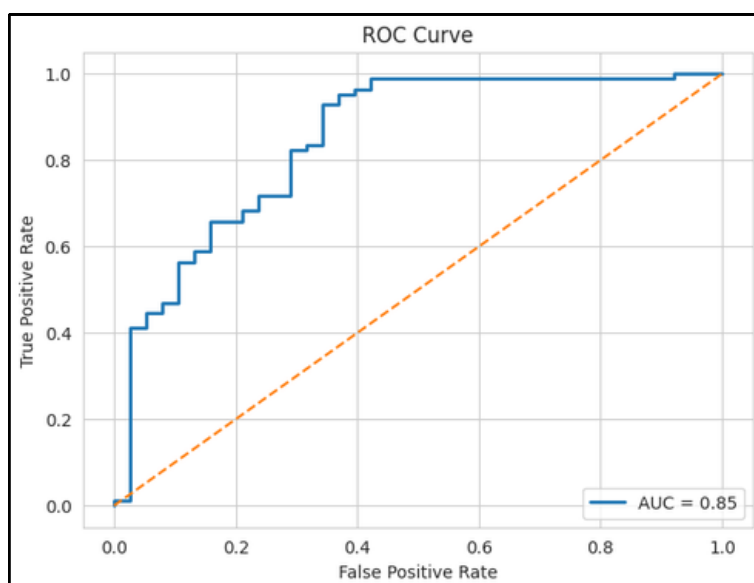


- The confusion matrix shows that the model correctly classified 84 approved and 22 rejected loan applications, with only a few misclassifications.
- This indicates that the model performs particularly well in identifying applicants likely to receive loan approval while maintaining relatively low classification error.

6.3.3 ROC curve

Figure 6.4

ROC Curve of Logistic Regression Model(Best Model)

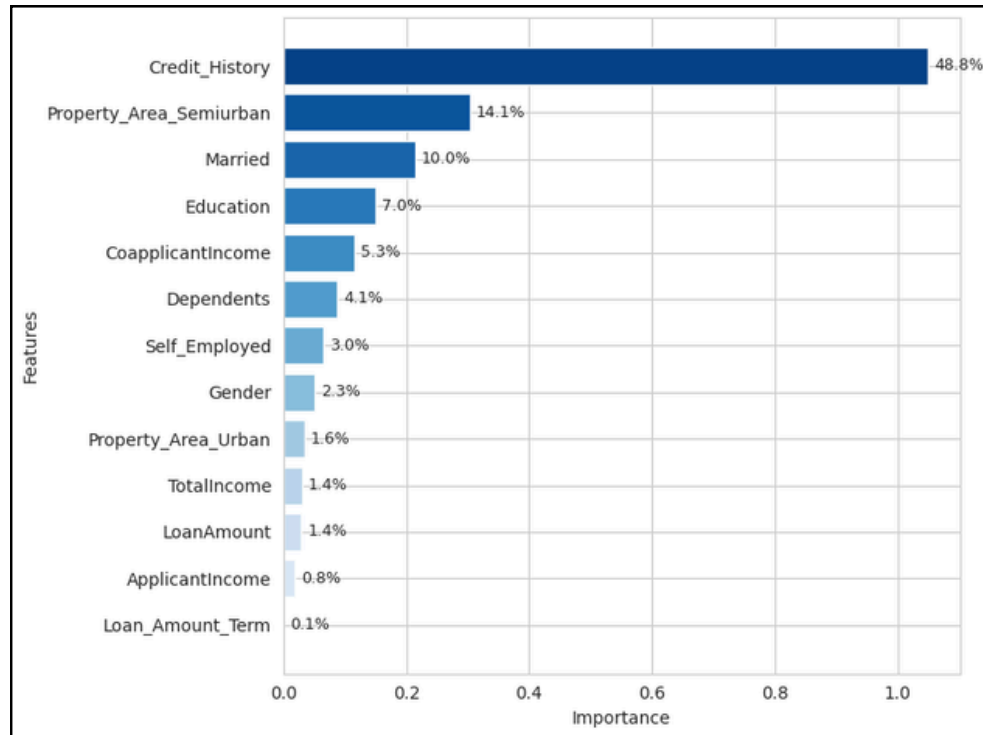


- The ROC curve was used to evaluate the classification ability of the Logistic Regression model.
- The model achieved an ROC-AUC score of 0.85, indicating good classification performance.
- Since the ROC curve remains well above the diagonal reference line, the model performs better than random classification.

6.3.4 Feature Importance

Figure 6.5

Feature Importance of Logistic Regression Model(Best Model)



- This analysis was conducted using the coefficients of the Logistic Regression model to identify the variables that most strongly influenced loan approval decisions.
- The analysis revealed that Credit_History was the most important predictor of loan approval, indicating that applicants with strong credit histories were significantly more likely to receive loan approval. Other important variables included Property_Area_Semiurban, Married, Education and CoapplicantIncome.
- These findings suggest that financial credibility and applicant characteristics influence loan approval decisions and also support the findings observed during the EDA.

6.4 COMPARISON BETWEEN EDA AND ADVANCED ANALYSIS RESULTS

- The findings observed during the Exploratory Data Analysis (EDA) were supported by the machine learning analysis results.
- During EDA (Figure 5.6), applicants with valid Credit_History showed much higher loan approval rates. The feature importance analysis (Figure 6.5) also identified Credit_History as the most influential variable in the Logistic Regression model.
- Variables such as Education, Married and Property_Area also showed visible relationships with loan approval during EDA and were later identified as important predictors in the advanced analysis.
- Histogram and boxplot analyses showed skewed distributions (Figure 4.2) and outliers in variables such as ApplicantIncome and LoanAmount. These findings supported the preprocessing decisions used during data preparation.
- EDA also showed that the dataset had only a moderate class imbalance. Despite this, the Logistic Regression model achieved strong classification performance without using balancing techniques such as SMOTE or undersampling.
- Overall, the consistency between EDA findings and machine learning results increased the reliability and validity of the study.

Summary of Advanced Analysis

The advanced analysis showed that Logistic Regression was the most suitable model for loan approval prediction among all evaluated machine learning models. The model achieved strong Accuracy, F1-Score, and ROC-AUC results, while the classification report, confusion matrix, and ROC curve confirmed its reliability and predictive capability. Feature importance analysis identified Credit_History as the strongest predictor of loan approval. Overall, the advanced analysis demonstrated that machine learning techniques can effectively support accurate and reliable loan approval prediction.

7. GENERAL DISCUSSION AND CONCLUSION

7.1 DISCUSSION

The main objective of this study was to identify the important factors affecting loan approval and to develop a machine learning model capable of predicting loan approval status accurately. Based on the results obtained from exploratory data analysis, preprocessing, model development and evaluation, the objectives of the study were successfully achieved.

- The exploratory data analysis (EDA) helped identify several meaningful relationships within the dataset. Variables such as Credit_History, Education, Married, Property_Area and income-related features showed noticeable effects on loan approval decisions. In particular, applicants with valid credit histories had a much higher chance of receiving loan approval compared to applicants without valid credit histories.
- Several machine learning models were evaluated, including Logistic Regression, Decision Tree, Random Forest, Gradient Boosting and Support Vector Machine (SVM). Among these models, Logistic Regression showed the best overall performance with strong Accuracy, F1-Score and ROC-AUC results. Although Decision Tree achieved a similar accuracy value, Logistic Regression demonstrated better classification balance and more reliable prediction capability.
- The model evaluation results further confirmed the effectiveness of the selected model. The classification report, confusion matrix and ROC curve analysis showed that the Logistic Regression model was able to predict loan approval status accurately and reliably.
- Feature importance analysis revealed that Credit_History was the most influential variable affecting loan approval decisions. Other important variables included Property_Area_Semiurban, Married, Education and CoapplicantIncome. These findings are consistent with real banking practices, where financial reliability and applicant background are important in loan approval decisions.
- An important strength of the study was the consistency between the EDA findings and the advanced machine learning analysis. The patterns observed during EDA were further supported by the feature importance analysis and model results, increasing the reliability of the study findings.
- The dataset had a slight class imbalance, with about 68.7% approved loan applications. Since the imbalance was not severe, techniques such as SMOTE or undersampling were not used. The selected model still achieved good performance, indicating that the imbalance did not significantly affect model reliability.
- Hyperparameter tuning using GridSearchCV also contributed to improving model performance and stability by identifying the most suitable parameter combinations for each algorithm. This allowed a fair comparison among models and improved the reliability of the final results.

7.2 CONCLUSION

- This study successfully identified the important factors affecting loan approval decisions using exploratory data analysis and machine learning techniques.
- The EDA findings revealed that variables such as Credit_History, Education, Married, Property_Area and income-related features had noticeable relationships with loan approval status.
- Several supervised machine learning models were developed and evaluated, including Logistic Regression, Decision Tree, Random Forest, Gradient Boosting and Support Vector Machine (SVM).
- Among all evaluated models, Logistic Regression achieved the best overall performance with an Accuracy of 86.18%, F1-Score of 0.908 and ROC-AUC score of 0.848.
- Feature importance analysis identified Credit_History as the most influential variable affecting loan approval decisions, followed by variables such as Property_Area_Semiurban, Married, Education and CoapplicantIncome.
- The classification report, confusion matrix and ROC curve analysis confirmed that the selected Logistic Regression model achieved reliable and effective predictive performance.
- The consistency between the EDA findings and advanced machine learning analysis increased the reliability and validity of the study results.
- Overall, the study demonstrates that machine learning techniques can effectively support automated loan approval prediction and assist financial institutions in making faster, more consistent and data-driven lending decisions.
- Future studies may improve prediction performance further by using larger datasets, additional financial variables or more advanced machine learning approaches such as ensemble and deep learning methods

8. REFERENCES

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